18.Implement image cropping, copying and pasting to select a region of interest (ROI) from the source image using Open CV.

Codec :

```
import cv2

# Read the input image
img = cv2.imread(r"C:\Users\ROHIT\Pictures\Instagram\IMG_20220920_205353_945.jpg")

if img is None:
    print("Error: Image not found")
else:
    # Select ROI: x, y, width, height
    x, y, w, h = 50, 50, 200, 150

    # Crop and copy the ROI
    roi = img[y:y+h, x:x+w].copy()

    # Paste ROI at a new location
    result = img.copy()
    new_x, new_y = 250, 100

    # Check whether the ROI fits in the image
    if (new_x + w <= img.shape[1] and
            new_y + h <= img.shape[0]):
        result[new_y:new_y+h, new_x:new_x+w] = roi

        cv2.imshow("Original Image", img)
        cv2.imshow("Cropped ROI", roi)
        cv2.imshow("Copied and Pasted Image", result)
```

```
        cv2.imwrite("roi_result.jpg", result)


        cv2.waitKey(0)
        cv2.destroyAllWindows()
    else:
        print("Error: Paste location is outside the image")
```

output:

